Climate Policy and the Ramsey Formula: Normative Justification, Underspecification, and Disaggregation

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Abstract

Welfare economics is inherently normative, both in its implications for policy and in the justifications of its parameters. This dissertation examines these normative dimensions through the Ramsey formula as applied to climate change, focusing on the ethical basis for its parameters and aggregation choices. Two central arguments are developed. First, the 'quasi-paradigmatic' status of welfare economics consequential good hinders substantive normative debate over the elasticity of marginal well-being, η, since the absence of a shared definition of well-being leaves competing positions talking past one another. Second, aggregation practices in climate policy modelling often obscure intra-generational inequalities, particularly between climate-safe and climate-affected groups. To address this, I propose a modification to the Ramsey framework that disaggregates these groups before applying the welfare transformation, making inequality impacts explicit while retaining methodological continuity.

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1 Introduction

Climate change presents an existential challenge, with impacts unfolding over decades and centuries. Decisions made today will shape the lives of future generations. In economic evaluation, the dominant framework for how we value future generations is encapsulated by the Ramsey formula (Ramsey 1928):

$$r = \delta + \eta g(C_t) \tag{1}$$

This expresses the discount rate on future consumption as a sum of three variables; the pure time preference (δ), plus the product of forecasted growth of consumption ($g(C_t)$) and the elasticity of well-being (η). Small shifts in any of these parameters can produce vast differences in climate policy guidance.

These parameters are inherently normative, embodying commitments about intergenerational and intra-generational justice; namely, how future individuals are valued relative to the present. This dissertation explicitly engages with this normativity.

Section 2 introduces inter-temporal welfare economics and argues for the normative justification of the Ramsey formula's parameters in climate policy. Section three examines the δ and η parameters, arguing for zero pure time preference and highlighting the difficulties in justifying a precise η value. Section 4 considers $g(C_t)$ and the ethical implications and justifications of aggregation, proposing a disaggregated model for climate-affected and climate-safe groups.

2 Normative economics

Normative economics is concerned with the evaluation of different economic states of affairs, processes and institutions. These evaluations define what 'should' be done in economics, (Hausman, McPherson, and Satz 2016, p.13). Much of normative economics operates under a quasi-paradigmatic assumption that some consequentialist metric or 'good' serves as the ultimate object of concern and optimisation. This notion judges an action or policy solely by how much it produces of this 'good' as a consequence, i.e., what should be done is the action or policy which produces the most of this 'good'. In normative economic models, this metric is typically assumed to increase with consumption, a relationship that functions as a low-level generalisation anchoring the logic of these models. A low-level generalisation refers to an empirical regularity described without reference to an underlying mechanism. For instance, Boyle's law describing observable volume and temperature, without appealing to the kinetic theory of gases (example borrowed from Ryan 1970, p.77).

While the terminology varies across literature, with terms like utility, well-being and welfare being commonly invoked, the underlying objective is the same: to maximise this measure of value. This consequentialist metric is understood and presented in different ways within the literature, of which I will briefly present now. Often, there is little consensus on the definitions of the specific terminology for this 'good', let alone a consensus on which should be used in normative economics. What follows is not an attempt to resolve these ambiguities, but rather to present some different conceptions in welfare economics to understand the differences.

- Subjective well-being: A person's own assessment of how well their life is going, which can include feelings, experiences and overall life satisfaction (Hausman, McPherson, and Satz 2016, ch.7).
- 'Objective' well-being: How well a person's life is going based on factors that are valuable independently of feelings or preferences,e.g., good health, education, meaningful work, social

relationships. There elements are treated as intrinsically good (Hausman, McPherson, and Satz 2016, p.130).

- Felicity: A momentary or instantaneous hedonistic flow of pleasure or pain (Dasgupta 2008, p.143). First introduced by Bentham in his felicific calculus (Bentham 1780, ch.4)
- Utility: A numerical index that represents the ranking of an individual's preference. As a formal theory of welfare, it does not specify what is intrinsically good for people, but how to find it out (Hausman, McPherson, and Satz 2016, p.119).
- Capabilities: Sen (1999) focused on the freedom and capabilities of individuals to live a life they have reason to value. This is instead of a bundle of goods and services they consume.

This lack of consensus as to what exactly entails this underlying 'good' is the reason for its role as a 'quasi-paradigm'. Kuhn describes a 'paradigm' as the "constellation" of theoretical assumptions, standard practices and methods, exemplars, etc.. which direct scientists in their work (Thomas Samuel Kuhn 1962, p.175). In short, it provides a model of the 'story' that, as scientists, we ought to tell, thereby providing a focus for research and experiment which would otherwise be unavailable. In absence of this, we would be left rudderless in a sea of undifferentiated data and models without some model to rule in relevant research and rule out the rest (Ryan 1970, p.80-81).

A widely accepted theory or assumption of this consequential 'good' would provide this basis for normative economics as a high-level generalisation or law. A high-level generalisation is a theoretical explanation of the underlying activity for a low-level generalisation (Ryan 1970, p.80). It allows for more rigorous research and direction for normative experimentation and debate, in the same way that the kinetic theory of gas provides a framework and direction of research for understanding Boyle's law; as particles in motion.[1] Without this clear theory or consensus, normative economics remains quasi-paradigmatic, relying on a shared heuristic of many overlapping and loosely held assumptions and modelling practices held together by this undefined consequentialist 'good', rather than a truly unifying paradigm. In short, there is no agreed upon high-level explanation for the low-level regularity being posited between consumption and this 'good'. This discussion, along with its implications for philosophical debate, will be explored further in a later section of this paper.

For clarity and consistency, I will refer to this underlying 'good' as social welfare or well-being throughout the remainder of this paper, in line with the common nomenclature of framing normative economics as welfare economics (Hausman, McPherson, and Satz 2016, p.14). This terminological choice is made solely for the sake of convenience and should not be taken to imply any substantive commitment to a particular theoretical account of this consequential good.

2.1 Evaluation across time

With this understanding, let's turn to the practice of modelling in welfare economics. *Social welfare*, W, is the fundamental metric to which every judgement in welfare economics ultimately appeals to and aims to maximise. W is the aggregation of each person's well-being, w, in the population. Assume that each person's well-being is derived purely from consumption, represented by the function $w = U_i(c_i)$, for an individual i. In a population of size N, where individuals are indexed by i ($i = 1, 2, 3, ..., N$), the consumption of individuals is aggregated into C_i. Using the function that

[1] The distinction between low- and high-level generalisations in normative economics is weaker than in Boyle's law: the link between consumption and a consequentialist "good" is already partly theory-laden, though less so than a full explanatory theory.

represents the relationship between consumption and well-being, this gives us the social welfare function, $W = U_i(C)$.

When applied to inter-temporal settings (i.e. decisions whose consequences extend across multiple generations), time is noted to be discrete and noted by t ($t = 0, 1, 2, 3, ...$), with the present being denoted as t=0. This gives us the social welfare function across time:

$$W_t = \Sigma_t \, U_i(C_t) \tag{2}$$

W_t is the social welfare of all generations derived by summing up each generation's social welfare, $U_i(C_t)$. The C_t for each future generation is assumed to be nothing more than forecast of future consumption, based on society's reading of technological process, government policies etc...

When evaluating intergenerational well-being at t=0, in the present, the social evaluator can choose to introduce a *pure time preference*. To do this in the evaluation of well-being at t=0, you add a discount rate to the social welfare of future generations:

$$W_0 = U(C_0) + U(C_1)/(1+\delta) + ... + (C_t)/(1+\delta)^t + ... =_{t=0} \Sigma[U(C_t)/(1+\delta)^t] \tag{3}$$

What the numerical value of δ should be is a heavily debated topic in the literature.

The other heavily debated topic in the literature is the relationship between consumption and well-being, namely the function $U_i(C_i)$. This function is defined by the numerical value assigned to the elasticity of marginal well-being, η:

$$U_i(C) = C^{1-\eta}/(1-\eta) \tag{4}$$

This relationship is defined by its marginal value:

$$U'(C) = C^{-\eta} \tag{5}$$

The number assigned to η governs the curvature of the well-being function; a larger η implies a more pronounced concavity. This means that each additional unit of consumption yields progressively smaller additions to well-being. As an *inequality aversion* parameter, a higher η increases the effect that this diminishing marginal welfare effect has. What the numerical value of η should be is the cause of this disagreement.

by using expression (2) we obtain [2];

$$1 + r = (1+\delta)(1+g(C_t))^\eta \tag{6}$$

When δ and $g(C_t)$ are both 'small' this can be approximated to the Ramsey formula for the social discount rate of consumption using logarithms:

$$r \approx \delta + \eta g(C_t) \tag{1a}$$

[2]Proof from Dasgupta (2008, p.149):

$$\begin{aligned}
\frac{1}{1+\rho} &= \frac{1}{1+\delta}\,\frac{U'(C_{t+1})}{U'(C_t)} \\
&= \frac{1}{1+\delta}\left(\frac{C_{t+1}}{C_t}\right)^{-\eta}\left[U'(C) = C^{-\eta}\right] \\
&= \frac{1}{1+\delta}\left[1+g(C_t)\right]^{-\eta}\left[C_{t+1} = (1+g(C_t))\,C_t\right] \\
&\implies \quad 1+\rho = (1+\delta)\left[1+g(C_t)\right]^{\eta}
\end{aligned}$$

This formula gives us an understanding of the rate at which future consumption is discounted in welfare economics, derived from the different numerical values in social welfare maximisation. Recall, welfare economics evaluates different states of affairs to determine which course of action we *ought to take*. This is one sense Ramsey (1928) intended when stating that r and the parameters determining it be normative; they fix the moral weight to be given to the welfare of future people when deciding what decision *should* be taken that extends into the future. Furthermore, the Ramsey formula portrays the way in which the social discount rate of future consumption is not a primary ethical object, that is, not a normative parameter fixed by moral judgement (Dasgupta 2008, p.150). It is derived from our conception of 1. intergenerational well-being, 2. the relationship between well-being and consumption and 3. estimated future consumption, a descriptive parameter.

2.2 Descriptive or Prescriptive?

How should these values be chosen? In the literature, two broad methodological approaches are commonly distinguished: 1. the descriptive method, 2. the prescriptive method.

- The descriptive method: Sometimes called the positive method. The social discount rate should be derived from the observed discount behaviour in society. The observed market interest rates and the preferences and values that individuals reveal in their consumer decisions inform this decision (Kelleher 2017, p. 442)
- The prescriptive method: Sometimes referred to as the ethicist method. The social discount rate should be chosen on the basis of philosophical argument concerning intergenerational ethics (Kelleher 2017, p.443).

Although the descriptive method starts from positive facts about markets and the prescriptive method starts from philosophical reasoning, both ultimately supply the same thing: a normative weight on future well-being. The use of descriptive numerical values doesn't entail a category change from normative to descriptive, but a methodological difference within the ethical weighting process. Therefore, the descriptive method must be defended on ethical grounds (a point emphasised by Posner and Weisbach 2010).

Before setting out my contention that the descriptive discounting approach is ill-suited to welfare-economic analyses of climate change, I first outline how and why observed behaviour in the market reveals a discount rate. In short, market prices provide us with information about the values that people in the market attach to different commodities. An example from Broome (1994, p.137):

- Imagine treating 'a bottle of wine delivered exactly one year from today' as a product in its own right, complete with its own price quoted today. Because that wine is not available for immediate enjoyment, it is called a dated commodity, i.e., its identity is tied to a future date, with a *present price*. Now compare the prices of two goods you could buy today: 1. a present commodity: a bottle of wine you can purchase and uncork right now, 2. the dated commodity: an identical bottle of wine that will arrive at a future date, at a price agreed today, a forward price. The percentage gap between these two prices is the bottle's discount rate. It captures the amount the individuals in the market value deferred consumption of the wine compared with immediate possession.

Dated commodities usually trade at a lower price than the identical good available today, for two main reasons:

1. Productive fertility: Broome (1994, p.139-140) begins with the observation that economies are fertile; whether fermenting grapes or running factory machines, you can expect to harvest more than one unit of that same resource (or close substitutes) in the future, depending on how much you invest and consume now. If you choose to invest instead of consume now, next year's increase will exceed the decrease in consumption this year. A consequence of this is that next years commodities must be cheaper than this year's. No one would be able to sell next year's bottle of wine today unless the present price of that dated bottle is low enough to beat this return from simply investing the wine (or its monetary value) in production. In this equilibrium, the gap that emerges in the price is grounded in the economy's technological fertility, not psychology. In a scenario where every consumer were perfectly patient, technological fertility would force a positive discount on dated goods. This effect is represented by η in the Ramsey formula; it represents the discounting effect of economic growth.

2. Imprudence: This observation is that of psychology. A person is imprudent when they do not attach the same value to their future well-being as they do to their present well-being (Broome 1994, p.154). Individuals in the market often discount their own future well-being, not for uncertainty or for the effect of productive fertility but due to their own myopia. This effect will undeniably be reflected in actual prices, so any descriptive account of actual prices will contain this effect. This effect is a pure time preference toward present welfare and is represented by δ in the Ramsey formula.

The descriptive method of discounting is to take the market price descriptions of individual preferences caused by these mechanisms and to apply them to the methodology of normative welfare economics, which evaluates which decision should be taken.

This paper does not address the question of whether the descriptive approach is plausible in general, but only as it is applied to climate economics. The first objection to the descriptive method of discounting is the very imprudence discussed earlier. To embrace the descriptive method does more than endorse a pure time preference in welfare economics. It also treats the psychological impatience underpinning this preference as a legitimate rationale and assumes that the exact numerical effect of imprudence should be used as the parameter for the pure time preference. However, when appealing to the fundamental metric of social welfare W_t, importing this psychological bias means systematically undervaluing long-term welfare and therefore not properly maximising W (Broome 1994, p.151-154), all underpinned by our own myopia. Broome (1994, p.132) acknowledges that the descriptive method is often just a 'practical' and easy way to get a discount rate which does not require "difficult judgments about well-being and how to aggregate it". While this may work as a practical shortcut for shorter time horizons, climate change policy spans extremely long time horizons, with some actions producing benefits decades or even centuries into the future. As a result, even a small difference, caused by imprudence, can lead to vastly different outcomes because of compounding effects. This difference will quickly compound into a magnitude that exceeds a manageable margin of error. Note that this critique is separate from my later challenge to pure time preference; it is directed solely at the descriptive approach.

The second objection to the descriptive method is also related to the long-time horizons involved in climate change policies. It is the problem of 'disenfranchised generations' (Broome 1994, p.151-154). Climate change policies extend far into the future, requiring us to to understand, evaluate and discount the welfare and consumption of many generations to come. A descriptivist approach, by definition, involves observing and applying the discount behaviour of only those *currently* in the

market towards consumption and well-being. The values of the future generations, who are having their welfare discounted, are entirely excluded from this method of discounting. Consequently, the benefits from climate policies that will mainly accrue to those future people, which require us to bear costs now, end up being systematically undervalued, even without the psychological bias of imprudence. This entails an exclusion of the very people the analysis is meant to protect.

My final objection to the descriptive approach in climate change discounting is the issue of *intragenerational disenfranchisement*. Markets are often praised for 'revealing' societies' values, yet they do so in proportion to purchasing power, not moral worth. Poorer households and areas, who often face the sharpest climate damages, possess so little purchasing power that their marginal willingness-to-pay barely registers in global markets. Because market prices are determined by dollars rather than people, the market will find its equilibrium in proportion to the richer individuals preferences who, most often, will be less climate affected (Chancel et al. 2021). The input of the poorer are therefore 'disenfranchised' in exactly same way that Broome (1994) describes for future people; their values on discounting will be systematically understated. The problem of intragenerational disenfranchisement will assume greater significance in subsequent sections.

This leaves us in the position that we cannot put 'our faith' in the descriptive method of discounting for evaluating climate change effects (Broome 1994, p.156) [3]. The objections put forward by Broome (1994) are simply two sides of the same coin. The long time period of climate change means that 1. the imprudence error size compounds and, 2. future individuals are systematically not taken into account. This leaves the prescriptive account; the values should be derived normatively. This is the second and more significant way in which Ramsey (1928) intended when stating that the numerical value r be normative (recall the first way; r is used in determining normative economic decisions). The values determining r should be based on philosophical justifications regarding what should matter in intergenerational well-being, not on observed behaviour. Evidently, the consumption forecast remains an empirical value, and the prescriptive approach does not entail basing this forecast on argumentation, only the η and δ parameters.

3 The Ramsey Formula

It has now been established that the social discount rate, r, and the parameters it is derived from are normative in two dimensions. These parameters spell out how we *should* discount intergenerational well-being and consumption and, on that basis, which economic choices we *should* make. The following sections examine these parameters and their role in greater detail.

It is now important to elucidate the distinction between the discounting of well-being (our fundamental metric) and the discounting of consumption. The discounting of well-being is best understood by revisiting Eqs. (2) and (3). Equation (2) defines the social welfare function across time as the sum of each generation's welfare. Equation (3) extends this by introducing the factor $1/(1+\delta)^t$ which falls geometrically with time, t. Any positive value of δ, the pure time preference, therefore assigns systematically *smaller weights to the well-being* of future generations; it discounts the well-being of future generations in the calculation.

In contrast, there is the discounting of consumption. This is represented by the η parameter in the Ramsey formula. η defines the relationship between observed consumption and social welfare, represented in equations (4) and (5). This parameter essentially 'translates' values of future con-

[3] The descriptive approach may still remain useful for evaluating policies which do not have long time horizons and do have a more homogenous population, two pertinent features of climate change policies

sumption and welfare into equivalent values of present consumption (Goulder and Williams 2012, p.4). A higher η means that, as consumption increases, an *extra unit of consumption yields less additional well-being.* For this reason, the η parameter is referred to as the index of inequality aversion because it reflects ethical judgments on the amount consumption should contribute to social welfare, *'be they in the same or different periods'* (Dasgupta 2008, p.150). Different philosophers and economists have argued for different numerical values for the parameter η and δ, but, the key point is that there is no contradiction in employing a high η while keeping a pure time preference of $\delta = 0$ because they are fundamentally discounting different aspects of welfare economics for different reasons.

Together with the descriptive parameter $g(C_t)$, δ and η determine the social discount rate on future consumption, r. A positive pure time preference, $\delta > 0$ increases the social discount rate on consumption. The effect of the inequality aversion parameter, η, on the social discount rate depends on the descriptive parameter $g(C_t)$. A higher numerical value of η increases the social discount rate *if and only if* $g(C_t) > 0$. It is commonplace in the economic literature to assume that economic growth will increase indefinitely (Dasgupta 2008, p.149), thus implying that η has the same effect on the social discount rate as δ; future consumption will be valued less than present consumption in welfare calculations. For this reason, a positive δ is often associated with a higher level of inequality aversion, because it discounts the future generation, who are wealthier, in the ethical calculus. However, if $g(C_t) < 0$, the two normative parameters, δ and η, assume diametrically 'opposite roles' (Dasgupta 2008, p.150). In contrast to η, the positive value of δ would continue to increase the social discount rate, creating an ethical preference for greater inequality in consumption, as future generations are given less weight while they become poorer in consumption. Understanding this difference becomes particularly significant when discussing climate change, which promises to negatively affect a significant number of individuals.

Below are some of the preferred values for the consumption and well-being discount in the climate change literature:

- Cline (1992): $\delta = 0$, $\eta = 1.5$
- Nordhaus (1994): $\delta = 3\%$ a year, $\eta = 1$
- Stern (2007): $\delta = 0.1\%$ a year, $\eta = 1$

The authors have derived these using the prescriptive approach to discounting, meaning these parameter values are derived from philosophical and ethical justifications. In the following sections, their arguments will be presented and scrutinised. A short clarification might be helpful here. The difference between these values may not seem significant, only a few percentage points. But, as previously mentioned, in the long run, even small differences in discount rates mean large differences in the welfare analysis (Dasgupta 2008, p.151). For illustration, consumption 100 years from now will be worth 17 times less in Nordhaus (1994) compared to Stern (2007). These impacts exacerbate the differences in the philosophical justifications within the literature.

3.1 A Pure Time Preference?

The clear outlier in these authors is Nordhaus' choice of pure time preference, δ, being the only author to include one.[4] Why do Nordhaus and other authors (including Heath (2017) and Arrow

[4]Stern's $\delta = 0.1\%$ a year only reflects uncertainty *only.* It is clearly delineated in his argument that he is not including a pure time preference on future well-being.

(1999)) include and argue for a discount on future well-being when making normative decisions? While it has been established that as a descriptive fact, individuals show imprudence and this manifests as pure time preference, it does not normatively justify its inclusion in climate change welfare evaluations. Nordhaus and others likewise reject mere imprudence as a justification for a pure time preference, but defend its inclusion on explicitly philosophical grounds; they too are prescriptivists.

Kelleher (2017) posits that this disagreement within the climate change welfare economics literature is due to a difference in the evaluative scope that different authors attach to the outcomes defined by the Ramsey formula. Although all authors agree that inter-temporal welfare economics is normative, in that it defines what should be done and distinguishes better from worse outcomes, they disagree on the way the Ramsey formula is taken to be normative. This difference in pure time preference can be defensible depending on the scope adopted (Kelleher 2017, p.448). Kelleher distinguishes two types of welfare outcome rankings based on their evaluative scope;

- Value rankings: These rankings evaluate different states of affairs in terms of impersonal value, i.e., well-being only. It is normative in the sense that different outcomes are only ranked based on how much 'they improve the world' without consideration for rights, justice or fairness (Kelleher 2017, p.450). These rankings are impartial with respect to well-being meaning they evaluate rankings *solely* in terms of well-being. With the exception of Ramsey, authors constructing these rankings acknowledge that these considerations are important but maintain that they lie outside of the evaluative scope of these rankings (Kelleher 2017, p.450-451).

- All-things-considered rankings: These rankings include any agent-relative obligations, as well as impartial value. These are any other types of moral reasons, including rights, duties and agent-relative obligations, e.g., the duty a government has to its own citizens, which can outweigh the impartiality of well-being . In contrast to value rankings, this ranking aims to be a definitive framework for determining action in policy decisions, i.e., they determine *all things considered* what ought to be done (Kelleher 2017, p.451-455).

Broome (1994, p.51) illustrates the distinction. killing a healthy patient to save five others would be unjust, yet might "improve the world" by increasing overall welfare. A value ranking adopts this latter view while acknowledging the justice concern, whereas an all-things-considered ranking gives priority to saving the patient.

It becomes quite clear through this elucidation which evaluative scope would not permit the inclusion of a pure time preference. No ranking can appeal to impartiality with respect to only well-being and include any obligations that can outweigh well-being. For this reason, the justification of a pure time preference relies on the defensibility of the all-things-considered evaluative scope of welfare economics. Before critiquing this approach, I will first outline a few of the arguments offered in support of including a pure time preference.

For institutional reasons, Heath (2017) supports the use of a pure time preference. His case parallels the way we let geography structure duties of international aid: affluent nations are not required to sacrifice an unsustainable share of GDP for far-off generations, and likewise each generation need not bankrupt itself for individuals many centuries in the future. A modest pure time preference,i.e., $\delta > 0$ would function as a temporal border, assigning each generation an attainable goal for helping future generations (Heath 2017, p.459). Another common justification for a pure time preference in the literature is the argument from 'excessive sacrifice' (as in Nordhaus (1994)

and Arrow (1999)). The claim is that a zero pure time preference would force the present generation to divert an intolerably large share of resources to the distant future. To limit that burden, a pure time preference is proposed. Nordhaus puts it plainly, warning that a zero rate would place a heavy load on our shoulders and would drown one's generation in a 'sea of concern' (Nordhaus 2013, p.43). I will not contest these specific arguments. My claim is that, though ethically important, they ought to remain outside the scope of welfare-based rankings. I put forward 2 objections to all-things-considered rankings. One is contingent and other is non-contingent

In an all-things-considered ranking these important considerations must be settled into a single δ value. All obligations must be taken into account to provide a definitive action framework. Agent-relative obligations are stance-independent duties that bind a particular agent and address that agent's own conduct , for instance, a doctor's obligation to protect not kill a healthy patient, or a government's obligation to its own citizens. In this context, stance-independent means the duty does not depend on whether the agent wants it, chooses it or recognises it. Importantly, this does not necessarily provide anyone else a reason to support this action (Alexander and Moore 2016, ch.2). Authors who defend including a positive pure time preference often do so by giving a single agent-relative reason (e.g., the argument from excessive sacrifice (Nordhaus 2013), the institutional argument (Heath 2017) or even the argument from harm put forward by Dasgupta (2008), which differentiates between climate harm and mere failure to aid). Yet they give no principled method for combining and operationalising this one obligation with the many others in the literature and reality. These duties are directed to different individuals, in different time periods and in different geographical locations, and they will inevitably overlap and conflict in the absence of an extremely detailed rule-based system with priority rules and exception clauses for each obligation (Richardson 1990, p.309). Without such a system, overlapping obligations cannot be coherently combined. Furthermore, the duties are often role-differentiated and context-specific, reflecting the particular mandates of of different agents or institutions, for example, each government has an obligation to its own citizens above others (Richardson 1990, p.385-287). There is therefore no single, unified set of duties that could coherently underpin a uniform pure time preference in climate policy, even if there is complete agreement on the normative obligations different agents ought to have. Heath (2017, p.443), himself, does not actually specify a numerical value, only offering an justification for including one(Heath 2017, p.443). Others, like Nordhaus (2013) do, but this is introduced in an ad hoc fashion as a *plausible* numerical value, without a procedure for integrating the full range of agent-relative obligations. While an ad hoc number seems innocuous, the compounding effects of discounting mean that even small differentials in plausible δ translate into enormous differences in the long-term. This is the contingent objection.

While the previous objection provides practical difficulties to all-things-considered rankings, the more fundamental objection is conceptual: representing agent-relative considerations as numerical parameters as numerical parameters in a welfare function commits a category error. Conflating important agent-relative consideration with consequentialist welfare rankings obscures these two distinct normative frameworks. A uniform δ in the Ramsey framework is an agent-neutral, consequentialist tool for ranking states of affairs. By contrast, agent-relative obligations bind particular agents regardless of whether they produce better overall happiness. Agent-relative and agent neutral reasons cannot simply be weighted against each other 'without stripping the former of their distinctive character (Alexander and Moore 2016, ch.4). One might, following Nagel (1986), try to allocate them to different perspectives, or following Nozick (1974), treat agent-relative norms as side constraints. But such approaches simply do not yield a cardinal scale. Quite simply, there is no way of incorporating such obligations into δ without altering their nature. There is no value of

pure time preference, which is simply a discount on well-being, that could correctly represent the rights on the one healthy individual being killed for the benefit of any number of unhealthy patients. Likewise, in aid choices, rich nations do not apply a compounding numerical discount to the welfare the more geographically distant the beneficiaries are; no obvious δ value would 'remove this heavy load of our shoulders' in a way which would respect the structural distinction of the agent-relative obligation of the institution. Therefore, the result is not a coherent 'all-things-considered' ranking, but a transformation of the important ethical duties into something they are not.

This leaves inter-temporal welfare economics with one option; it should act purely as a value ranking. Without the inclusion of agent relative considerations in welfare evaluations, the argument for a pure time preference essentially dissolves. The welfare sums are impartial with respect to welfare, meaning the weight of one's welfare is the same "regardless of the generation to which that person belongs" (Kyllönen and Basso 2017, p.17). Only the amount of welfare itself counts in these models, not the identity or time of the person experiencing it. This yields a social welfare function of the following form, without any pure time preference $\delta = 0$. Evaluating intergenerational well-being at $t = 0$ [5]:

$$W_0 = \Sigma_t \, U_i(C_t) = U(C_0) + U(C_1) + U(C_2) + U(C_t) \tag{7}$$

the Ramsey is now defined:

$$r = \eta \, g(C_t) \tag{8}$$

This represents the social discount rate, when it is completely impartial with respect to well-being.

3.2 η: Consumption and Well-being

The social discount rate is now fully determined by the normative parameter, η and the descriptive parameter, $g(C_t)$, which is a forecast of future consumption. Together, they define the social discount rate which distinguishes better from worse outcomes, solely in terms of well-being. The normative parameter, η, defines the relationship between consumption and well-being, presented in equation (4) and (5), and is referred to as the inequality aversion parameter. Recall, this parameter should be derived normatively. It should be based on philosophical justification for what the relationship between well-being and consumption should be.

I will begin by presenting the normative implications of alternative numerical values for η. Any numerical value $\eta > 0$ will have the effect of diminishing marginal well-being to consumption. This means that, as long as η is positive, the additional well-being derived from consumption falls as consumption rises. Both, Stern (2007) and Nordhaus (1994) propose a parameter value $\eta = 1$. This implies that *proportionate* increases in consumption have equal social worth across individuals and across time periods, whereas equal absolute increases do not. Cline (1992) and Dasgupta (2008) propose a value larger than 1, with Cline proposing a value of 1.5 and Dasgupta open to a value larger than 1.5. If η is greater than 1, diminishing marginal well-being is stronger, and a given percentage increase counts less at higher consumption. Conversely, a value smaller than 1

[5] Proof:

$$W_0 = \Sigma_t \, U_i(C_t) = U(C_0) + U(C_1)(1+0)^1 + U(C_2)(1+0)^2 + U(C_t)(1+0)^t = U(C_0) + U(C_1) + U(C_2) + U(C_t)$$

still holds diminishing marginal well-being, but a given percentage increase counts more at higher consumption. In the literature, there has been some experimentation with a non-constant η, which itself varies with consumption, for example in Gollier (2002). This could be used to accelerate the effect of diminishing marginal well-being at higher income levels. At this stage, it is important to note that no matter the value of η, if it is a constant or if it varies with consumption, as long as $\eta > 0$, the relationship between well-being and consumption will be concave. In short, well-being will always increase at a diminishing rate with consumption. The implications of this choice in η is that it will fix the value given to individuals depending on their real consumption level. The higher the numerical value for the η parameter is, the less weight given to the consumption of those at a greater consumption level, i.e., more averse to inequality. The numerical value assigned to η therefore effects which outcome is better or worse, in terms of well-being.

Let's turn to the normative justifications for the η values. One approach is through a sort of ethical calibration by studying consumption changes between individuals that are judged to be *ethically equivalent* by eq. 7 (Dasgupta 2008, p. 151). An example provided by Dasgupta (2008, p. 152):

- Individual 1 and 2 have identical $U(c_t) = W$ functions. At purchasing power parity, an individual 1 has an annual consumption of \$360 and individual 2 has an annual consumption of \$36,000. If $\eta = 2$, a 50% decrease in individual 2's consumption is ethically equivalent to a 1% decrease in individual 1's consumption. From there, one can 'calibrate' the η parameter until the trade-off ethically seems reasonable.

Dasgupta (2008, p.157) prefers a greater η value for its higher inequality aversion and its avoidance of unreasonable recommendation of policy action. Stern (2007, p.46) explicitly affirms the choice of the η parameter as a "value judgement" of inequality and introduces it in a similar manner to the 'ethical calibration' introduced above. Yet, in practice, when defining a value of $\eta = 1$, Stern (2007, p.161) employs an "empirical estimate" of the marginal utility of consumption, from Pearce and Ulph (1999) to determine his numerical value. In a similar vein, Cline (1992, p.253) links η to ethical views via the tax system. He notes that any genuinely progressive taxation implies an elasticity of marginal well-being greater than one. Nevertheless, as did Stern, Cline continues by citing econometric evidence of diminishing marginal utility, from Fellner (1967), which separates imprudence and isolates the diminishing effect which points to an effect around $\eta = 1.5$. Finally, it is worth noting that both Dasgupta (2008, p.166) and Stern (2007, p.50) acknowledge the absence of an *a priori* ethical justification for maintaining η as a constant. The assumption that η is independent from consumption only has "tractability" to commend it.

I have intentionally kept the presentation of the literature's normative justifications for the specific numerical value assigned to η brief for two reasons. First, although the authors are confident about including a diminishing consumption effect on well-being, they provide little transparent justification for preferring one precise value over another, for example why 1.5 rather 1.4 would be ethically superior. Second, as introduced at the beginning of this paper, welfare economics faces a foundational difficulty. The consequential good at its core is quasi-paradigmatic, making normative deliberation and comparison about its exact relationship with consumption impossible.

It is at this point that the discussion started at the beginning of this paper becomes pertinent. The theory of 'well-being' is a messy area in philosophy and economics, and it is even difficult to categorise various theories (Hausman, McPherson, and Satz 2016, p.119). This paper will make no attempt to undertake the colossal task of deciphering between these theories and offering a definitive answer. Instead, I will attempt to bring to light the impact of this quasi-paradigm on the

normative implications and justifications for η.

Let us begin with an analogy from classical physics: Before we had Newtonian laws of motion, there were Galilean laws and before that Aristotelian laws. Each set of laws provided a 'paradigm' for its time, shaping how people designed experiments and interpreted phenomena such as a ball falling from a one story building (Ryan 1970, p.66). Each set of laws was defined by a distinct theory or 'story' of an object's motion. For instance, Aristotle posited the assumption that bodies have natural terminal velocity. Newton rejected this idea, explaining the effect instead as a result of gravity and drag forces (Ryan 1970, p.65). Importantly, all three laws of motion are very accurate for the kinds of motion Aristotle could observe, i.e., bodies moving near the ground. His laws were later falsified by experiments in broader contexts beyond his reach, requiring a new paradigm for understanding these motions (Ryan 1970, p.66). Now imagine, for the sake of argument, we could build a time machine and put Newton, Galileo and Aristotle into a room and ask them to explain the motion of a tennis ball falling from the ceiling. They would be unable to reach an agreement, not because of a lack of evidence, but because each would interpret the evidence through an entirely different paradigm, even if all three of them can predict the balls motion with high degrees of accuracy.

In the same way, moral and evaluative debates hinge on foundational commitments; as Hausman, McPherson, and Satz (2016, p.99) observe, these "must be answered first, because everything else depends on their answers". In the case of η, a definitive conception of well-being is required to compare and debate the normative justifications for different numerical values. Stern (2007) offers a striking example of the difficulties that arise from an ill-defined conception. Like all authors, Stern defines a consequentialism framework. Yet, when defining this 'good' Stern appeals to various conception of well-being and Stern, himself, admits that this multi-dimensionality creates difficulties when trying to encapsulate them all into a single scalar, η (Stern 2007, p. 29). In his paper, Stern introduces "utility or welfare" (p.28) as deriving from a broad number of consumption goods, and gives an objective list including "education, health, the environment, and consumption" (p.29). They will be valued 'to the extent that individuals value the environment' (p.28) which will act as "both goals and instruments". Too derive his value of $\eta = 1$, Stern uses an empirical estimate, removing the effect of the psychological effect of imprudence. These all point to different understandings of well-being, some of which I introduced above; the objective list of outcomes (health, education, etc..), the satisfaction of preferences, a psychological understanding etc... To unify these understandings, Stern points to the understanding that 'in reality' many of these 'overlap' (Stern 2007, p.41). This outcome exemplifies the issue of a 'heuristic' quasi-paradigmatic consequentialism that informs the selection of η, rather than a definitive framework. This is not to say that these different conceptions of well-being are not justifiable, in fact leaving it ill-defined means Stern avoids strongly committing himself to one conception, whether substantive or formal. However, this lack of foundation makes normative justification and discourse difficult. Dasgupta (2008, p.143) leaves the definition quite broad, simply defining the consequential good, felicity, as "instantaneous utility which depends solely on his or her consumption level", somewhat assuming that the framework has been well-established. In contrast to Stern, Cline (1992, p.240) defines his consequentialist framework much narrower, being a self-ascribed 'welfarist'. This is usually, but not always, associated with the satisfaction of preferences, a formal theory of consequentialism, but even then, without a direct definition from Cline, we are somewhat left assuming this (Hausman, McPherson, and Satz 2016, p.218).

As philosophers of economics, we find ourselves in the same position as Aristotle, Galileo and Newton do in their room. While the literature is unified on the use of a discount on consumption,

any attempt to compare the normative justifications put forward for the specific η parameter by each author will be handicapped, as each is working in their own 'paradigm' or theory of happiness. Aristotle, Galileo and Newton while agreeing on the motion of the ball, are left unable to deliberate on predictions and will be left speaking past each-other about the theory behind the ball, and what might happen to it in hypothetical scenarios. Similarly many disagreements on the parameter value of η will be the case of authors speaking past each other, as their understanding of the consequential good will be slightly different, only unified by a heuristic understanding of many overlapping understandings. How can we deliberate in detail on the relationship between consumption and 'some consequentialist good/metric' when the definition of the good/metric is left ill-defined? This is a foundational problem in the choice of η beyond a rough estimate.

It may seem that I am calling for the solving of an insurmountable task, which would "send economists running back to their graphs" (Hausman, McPherson, and Satz 2016, p.119). If a unifying theory of happiness could have been found, it would've created this paradigm for welfare economics and normative discourse and experimentation could flourish. Nonetheless, it remains important to register this deep indeterminacy in the literature, since disagreements over η may often reflect unstated commitments about well-being rather than normative disagreements about inequality. As a partial consolation, even if no unifying theory can found, I have argued that these welfare rankings should not be providing a definitive ranking for action, but a ranking of better and worse outcomes in terms of well-being. Therefore, the consequentialist good need not be 'the definitive well-being', it needs only be explicitly defined to understand the normative implications and justifications of η. If Cline had defined welfarism, his $\eta = 1.5$ could be evaluated within a clear framework, allowing its impact on a specific happiness measure to be tested with counterfactuals.

For the remainder of this paper, I will remain as ambivalent as I was prior to this section. I will refer to this ill-defined consequential good as well-being or welfare, without committing to one (with due appreciation of the irony). The relationship between this consequential good and consumption is concave, reflecting the diminishing marginal well-being effect. Since I'm not defining this consequential good, I will not provide a value for η other than it being greater than 0, which is agreed upon within the literature.

4 The Descriptive Parameter

As outlined in the previous section, the social discount rate is fully determined by the normative parameter η, and the descriptive parameter, $g(C_t)$, which is the growth rate of a forecast of future consumption, which has been aggregated into one numerical value. The normative parameter is the numerical value assigned to the relationship between consumption and well-being, and in the previous section, I presented the problem that an ill-defined well-being has on understanding and comparing normative justifications for the value.

The descriptive factor $g(C_t)$ is the growth rate of consumption, which, in the simple model I presented above as eq. (7) , evaluates intergenerational well-being at t=0:

$$W_0 = \Sigma_t U_i(C_t) = U(C_0) + U(C_1) + U(C_2) + ... + U(C_t) \tag{7}$$

This model is simple for 2 major reasons.

1. There is no uncertainty modelled. Within the model we assume a deterministic picture, where the theory or model of the economy being used to determine future consumption will, without a doubt, happen (Dasgupta 2008, p. 158). In models which include the factor of uncertainty, a

$\delta > 0$ is often used to reflect the uncertainty of the existence of future generations. This is not in opposition to my argument against a pure time preference. One can give future well-being the same weight as present well-being, while acknowledging the uncertainty involved. This uncertainty manifests as $\delta > 0$ for *only* uncertainty reasons.

2. Each generations consumption is aggregated into one value to "bypass intragenerational issues", Dasgupta (2008, p.146). This allows just one value to represent a undeniably heterogeneous group, i.e., in each generation, the individual's consumption will not be equal.

When aggregated into one value, the normative implications of the $g(C_t)$ value are very clear. When $\eta > 0$, $g(C_t) > 0$ implies that *every* future individual's consumption will be valued less than present individuals, no matter if the individuals themselves consume less. Vice versa, if $g(C_t) < 0$, *every* future individual's consumption will be valued more than present individuals.

4.1 The underdetermination problem

In his essay 'Objectivity, Values and Theory choice' Thomas S. Kuhn (1977) lists five epistemic values that scientists, including economists, use to guide their judgements. This list is not exhaustive, but he puts them forward as being 'individually important' (Thomas S. Kuhn 1977, p.357). These 5 values are accuracy, simplicity, breadth of scope, internal and external consistency, and fruitfulness. Epistemic values are values that determine what constitutes sound scientific practice, and should be used to objectively define 'good scientific theory' (Thomas S. Kuhn 1977, p.357).

The philosophical significance of the aggregation choice lies in its epistemic underdetermination. In short, multiple choices about aggregation may satisfy different epistemic desiderata. Most notably, all aggregation choices will be equally accurate, as the underlying model or theory for the forecast of consumption will be the same. The decision to simply sum, to mean average, to median average, to modal average, to priority weight when averaging, or to disaggregate the consumption forecasts will each meet different standards of scientific adequacy.

Crucially, each decision encodes distinct frameworks for weighting each individual's consumption, a fundamental issue of distributive ethics. Different aggregation techniques will affect normative outcomes of the model in ways that extend far beyond their epistemic properties. For example, choosing to mean average versus median average each generation's consumption may change which policy outcome ranks as best or worst in terms of well-being. Yet, in relation to Kuhn's epistemic values, these approaches are functionally identical, both are simple, fruitful and allow for breadth of scope. This creates a gap where epistemic considerations alone cannot determine the methodological choice. When epistemic reasons fail to provide definitive guidance, ethical values necessarily step in to fill this gap.

This mirrors what Longino (1995) identifies in her analysis of theoretical virtues in science more broadly. Even seemingly neutral epistemic values like simplicity carry socio-political commitments. She compares Kuhn's virtue of simplicity with the virtue found in feminist philosophy of science, ontological heterogeneity (Longino 1995, p.387). Simplicity seeks order in otherwise confusing phenomena, while ontological heterogeneity treats the world's individual differences as a basic feature (Thomas S. Kuhn 1977, p.357; Longino 1995, p.387). Longino points to the fact that both serve particular functions despite appearing purely epistemic (p.388).

The aggregation problem in welfare economics exemplifies this. Beyond accuracy (what Longino refers to as empirical adequacy), the choice of different aggregation methods as well as the choice whether to aggregate cannot be resolved simply by appealing to epistemic values alone, because

these values themselves carry normative assumptions about scientific practice and potentially desirable social outcomes. Just as Longino shows that the value of simplicity in economic theory can encode assumptions about social hierarchy, for instance, by treating households as having a single decision-makers (p.393), the choice of aggregation methods in Ramsey formula welfare economics embed assumptions about whose consumption counts to deciding the social discount rate for the whole generations consumption.

4.2 Aggregation for Climate change

It is for this reason that, when Dasgupta (2008, p.146) states that aggregating the consumption of each generation to the average level 'bypasses intragenerational issues', this represents far more than a neutral technical simplification. This framing conceals a fundamental normative choice about whose consumption matters within each generation, which the underlying economic output does not inherently determine. Namely, those who consume the most will decide the social discount rate on the entire generation, a simple fact of a mean average.

This becomes problematic when applying this framework to climate change. A fundamental characteristic of climate change is that it affects people unequally, and often those most severely affected are the poorest and most vulnerable populations. There are multiple pathways for this disproportionate impact . The great majority of poorer people live in the geographical tropics, where the direct physical impacts of climate change are projected to be most severe (Sachs, Mellinger, and Gallup 2000, p.2). Poorer individuals depend more directly or indirectly on activities that may be most impacted by climate change, notably agriculture and fisheries (World Bank 2007, p.1). Rising sea-levels and changing precipitations patterns directly threaten food security, health and housing and the poorest will be the least able to adapt (Intergovernmental Panel on Climate Change 2022, p.25). An approach to aggregation that ignores these intra-generational effects would therefore be missing the fundamental distribution dimension of climate policy. Furthermore, richer countries, who dominate the mean average consumption (because they will swell the overall sum in calculation) will be affected far less by climate change (Callahan and Mankin 2022, p.7).

This is the problem of *intra-generational disenfranchisement* I mentioned in chapter 2.1. If we propose an aggregation approach which disenfranchises the poorer, and usually climate-affected, individuals, then no choice of η can remedy intra-generational injustice, because the harm stems from the aggregation rule rather than the parameter. Failing to account for intra-generational inequalities overlooks not only the harms caused by climate change but also the benefits of the policies being evaluated. Climate policies, typically classified as mitigation or adaptation, have uneven economic effects across different groups (Yamaguchi 2019, p.958).

More recently, there has been a growing literature on the intragenerational effects of climate change. For example, Yamaguchi (2019) has considered a consumption discount rate that can be decomposed into the growth effect,based on aggregate consumption growth and an intra-generational distribution effect, based on changes in consumption shares across groups. This means that even with no growth, changes in distribution can alter the social discount rate. Growth and inequality shifts can either offset or amplify each other (Yamaguchi 2019, p.961-963):

$$r(t) = \delta - \underbrace{\frac{1}{t}\ln\Big[\big(\tfrac{C(t)}{C(0)}\big)^{(1-\sigma)(1-\eta)-1}\Big]}_{\text{growth effect}} - \underbrace{\frac{1}{t}\ln\Big[\sum_{i=1}^{N}\phi_i(0)\Big(\tfrac{\alpha_i(t)}{\alpha_i(0)}\Big)^{(1-\sigma)(1-\eta)}\Big]}_{\text{distribution effect}} \tag{9}$$

The first term after δ captures the *growth effect*: aggregate consumption growth weighted by both η (elasticity of marginal utility) and σ (inequality aversion between groups). The second term is the *distribution effect*, which is positive if consumption becomes more equal across groups and negative if it becomes more unequal. $\alpha_i(t)$ is the share of total consumption of group i at time t, and $\phi_i(0)$ is its initial value share, combining this share with its marginal utility. Together, the two effects show that $r(t)$ depends not only on how much the economy grows but also on how that growth is shared.

Similarly, Azar and Sterner (1996) introduce equity weights when aggregating, by splitting the population into rich and poor. The marginal effect on well-being to consumption is then normalised to the richer group, giving the richer group a weight of 1 (w_{rich}) and the poorer one a weight of (rich income)/(poor income) ,w_{poor} (Azar and Sterner 1996, p.179). Changes in income of the population are weighted to the poorer group in the population.

$$\Delta c^{\text{EquityWeighted}}(t) = \Delta c_{\text{rich}}(t) + w_{\text{poor}}\, \Delta c_{\text{poor}}(t) \tag{10}$$

4.3 Disaggregating Growth

The aggregation methods discussed so far account for normative implications and justifications of climate change. Now, I propose a different modification to the basic model introduced in the first section, one which disaggregates consumption. I will first outline this disaggregation, with an illustrative example, and then compare it to other models that incorporate intra-generational well-being.

In this model, the population is divided into two groups; the climate safe group c_s and the climate affected group c_a. The climate safe group consists of individuals who are unaffected or minimally affected by climate change, meaning their consumption will continue to rise despite its impacts. The climate affected group consists of individuals whose consumption patterns will be adversely altered by climate change. These groups exhibit diametrically opposite social discount rates, since one group's consumption is increasing while the other's is decreasing. In this model, evaluating the consumption of well-being at t=0 is :

$$W_s = \Sigma_t\, U_i(C_{ts}) \tag{11}$$

(Eq) 11 represents the intergenerational sum of well-being for climate safe individuals.

$$W_a = \Sigma_t\, U_i(C_{ta}) \tag{12}$$

(Eq) 12 represents the intergenerational sum of well-being for climate affected individuals.

$$W_0 = W_s + W_a$$

Together these give the total welfare. In this disaggregated approach, the consumption trajectories of climate safe and climate affected groups are converted into well-being before aggregation. This contrasts with the models that first combine consumption, with inequality weights, and then apply the well-being transformation.

To illustrate the difference, I present an example in a simple two period model, with period i and period j. When each generation's consumption is aggregated the inter-temporal sum looks like this:

$$W = U(c_i) + U(c_j) \tag{12}$$

we determine policies based on the amount they change welfare:

$$\Delta W = -U'(c_i)\Delta c_i + U'(c_i)\Delta c_j \tag{13}$$

Suppose a climate policy benefits the climate affected group more. Under disaggregation, determining policies requires disaggregating consumption before transforming it into well-being. For the climate safe group:

$$\Delta W_s = -U'(C_{is})\Delta C_{is} + U'(C_{js})\Delta C_{ij} \tag{15}$$

For the climate affected group:

$$\Delta W_a = -U'(C_{ia})\Delta C_{ia} + U'(C_{ja})\Delta C_{ja} \tag{16}$$

The total policy impact is then:

$$\Delta W_s + \Delta W_a = \Delta W \tag{17}$$

When comparing the change in well-being for the *same policy*, $\Delta W_{disaggregated} > \Delta W_{aggregated}$. This is because the positive consumption effect on the climate affected group has a proportionally greater impact on overall well-being when the transformation is applied separately to each group's consumption path. This difference in outcomes follows from Jensen's inequality and the principle of diminishing marginal utility. Because the well-being function $U(\cdot)$ is concave, a given absolute change in consumption has a greater effect on well-being when it occurs at a lower consumption level. The climate affected group is poorer, so gains for them are valued more in well-being terms than equal-sized changes for the richer climate safe group.

When consumption is aggregated before transformation, the poorer group's consumption level is averaged with that of the richer group, muting this effect. By disaggregating first, each group's well-being is calculated at its own consumption level, ensuring that welfare calculations fully reflect the greater marginal value of changes for the climate affected group.

While the other models also focus on equitable outcomes, this disaggregation model provides a few advantages. Firstly, it champions impartiality in the treatment of well-being by applying the same η value to both intra and inter-generational consumption. By avoiding equity weights or additional distributional parameters, it ensures that the ethical judgement linking consumption to well-being remains consistent across all individuals and time periods. Second, it preserves normative transparency. The higher marginal value placed on gains to poorer, climate-affected groups is explicit in the calculation itself, rather than being embedded back into an aggregated growth term where its effect can be difficult to isolate or interpret. Third, it operates entirely within the familiar Ramsey framework, altering only the aggregation step. This makes it both simple to implements and methodologically continuous with the current literature, while still producing results that take into account the key feature of climate change; its inequitable effects.

5 Conclusion

The central contribution of this dissertation is to clarify the normative implications and justifications in inter-temporal welfare economics when applied to climate change. The need for normative justifications for the discount rates on well-being and consumption were highlighted, as the descriptive approach based on market observation implicitly imports human's psychological imprudence and disenfranchised individuals in the future, and current individuals with low purchasing power.

Welfare economics' role was presented as being impartial to well-being, therefore dissolving any argument for a time preference.

A major argument of this dissertation regarded the role of η in linking consumption to well-being. While consensus exists in the literature on the diminishing marginal value of consumption, the justification for a precise numerical value was shown to be undermined by the quasi-paradigmatic nature of welfare economics' consequential "good". Without an explicit and shared definition of well-being, debates over η risk talking past one another, making transparency in its definition essential for coherent normative evaluation. Looking ahead, a pressing task for future research is to identify and clearly define the consequential good at the heart of welfare economics. While perhaps unsurprising once brought into focus, this task is nothing less than foundational. Without such clarity, debates over η risk drifting in circles, with arguments talking past one another. Constructing a fully worked-out theory of well-being is a task of truly staggering proportions, perhaps beyond the reach of any single project, but that does not absolve authors from at least stating their own conception explicitly. Doing so would not only make their normative justifications for η more transparent, but also invite a more meaningful and productive dialogue across differing frameworks.

Finally, after establishing the implicit normative justifications embedded in aggregation choices, this paper presented an alteration to the model which disaggregates climate-affected and climate-safe groups before applying the welfare transformation, for intra-generational inequality reasons. This approach makes the distributional impacts of climate change explicit in the welfare analysis, and does so while remaining fully consistent with the Ramsey framework.

References

Alexander, Larry and Michael Moore (2016). "Deontological Ethics". In: *The Stanford Encyclopedia of Philosophy*. Ed. by Edward N. Zalta. Winter 2016. First published 2012. Metaphysics Research Lab, Stanford University. URL: http://plato.stanford.edu/archives/win2016/entries/ethics%2Ddeontological/.

Arrow, Kenneth J. (1999). "Discounting, Morality, and Gaming". In: *Discounting and Intergenerational Equity*. Ed. by Paul R. Portney and John P. Weyant. Washington, D.C.: Resources for the Future, pp. 13–21.

Azar, Christian and Thomas Sterner (1996). "Discounting and Distributional Considerations in the Context of Global Warming". In: *Ecological Economics* 19.2, pp. 169–184. DOI: 10.1016/0921-8009(96)00065-1. URL: https://doi.org/10.1016/0921-8009(96)00065-1.

Bentham, Jeremy (1780). *An Introduction to the Principles of Morals and Legislation*. Ed. by J. H. Burns and H. L. A. Hart. New York: Dover Publications.

Broome, John (1994). "Discounting the Future". In: *Philosophy & Public Affairs* 23.2, pp. 128–156. URL: http://www.jstor.org/stable/2265483.

Callahan, Christopher W. and Justin S. Mankin (2022). "Globally unequal effect of extreme heat on economic growth". In: *Science Advances* 8.43, eadd3726. DOI: 10.1126/sciadv.add3726.

Chancel, Lucas et al. (Dec. 2021). *World Inequality Report 2022: Executive Summary*. https://wir2022.wid.world/executive-summary/. World Inequality Lab, Paris School of Economics.

Cline, William R. (1992). *The Economics of Global Warming*. Washington, DC: Institute for International Economics.

Dasgupta, Partha (2008). "Discounting climate change". In: *Journal of Risk and Uncertainty* 37.2/3, pp. 141–169.

Fellner, William (1967). "Operational Utility: The Theoretical Background and a Measurement". In: *Ten Economic Studies in the Tradition of Irving Fisher*. Ed. by Unknown. New York: John Wiley & Sons, pp. 39–74.

Gollier, Christian (2002). "Time Horizon and the Discount Rate". In: *Journal of Economic Theory* 107, pp. 463–473. DOI: 10.1006/jeth.2001.2952.

Goulder, Lawrence H and III Williams Roberton C (Aug. 2012). *The Choice of Discount Rate for Climate Change Policy Evaluation*. Tech. rep. National Bureau of Economic Research. DOI: 10.3386/w18301. URL: http://www.nber.org/papers/w18301.

Hausman, Daniel, Michael McPherson, and Debra Satz (2016). "WELFARE AND CONSEQUENCES". In: *Economic Analysis, Moral Philosophy, and Public Policy*. Cambridge University Press, pp. 107–172.

Heath, Joseph (2017). "Climate Ethics: Justifying a Positive Social Time Preference". In: *Journal of Moral Philosophy* 14.4, pp. 435–462. DOI: 10.1163/17455243-46810051. URL: https://doi.org/10.1163/17455243-46810051.

Intergovernmental Panel on Climate Change (2022). "Summary for Policymakers". In: *Climate Change 2022: Impacts, Adaptation and Vulnerability*. Cambridge and New York: Cambridge University Press, 3 to 33.

Kelleher, J. Paul (2017). "PURE TIME PREFERENCE IN INTERTEMPORAL WELFARE ECONOMICS". In: *Economics and Philosophy* 33.3, pp. 441–473. DOI: 10.1017/S0266267117000074.

Kuhn, Thomas S. (1977). "Objectivity, Value Judgment, and Theory Choice". In: *The Essential Tension: Selected Studies in Scientific Tradition and Change*. Chicago: University of Chicago Press.

Kuhn, Thomas Samuel (1962). *The Structure of Scientific Revolutions*. Ed. by Otto Neurath. Chicago: University of Chicago Press.

Kyllönen, Simo and Alessandra Basso (2017). "When Utility Maximization is Not Enough: Intergenerational Sufficientarianism and the Economics of Climate Change". In: *Ethical Underpinnings of Climate Economics*. Ed. by Adrian Walsh, Susanne Hormio, and Duncan Purves. London and New York: Routledge.

Longino, Helen E. (1995). "Gender, Politics, and the Theoretical Virtues". In: *Synthese* 104.3, pp. 383–397. (Visited on 08/12/2025).

Nagel, Thomas (1986). *The View from Nowhere*. New York: Oxford University Press.

Nordhaus, William D. (1994). *Managing the Global Commons: The Economics of Climate Change*. Cambridge, MA: MIT Press.

— (2013). *The Climate Casino: Risk, Uncertainty, and Economics for a Warming World*. Yale university press.

Nozick, Robert (1974). *Anarchy, State, and Utopia*. New York: Basic Books.

Pearce, David and David Ulph (1999). "A Social Discount Rate for the UK". In: *Economics and the Environment: Essays on Ecological Economics and Sustainable Development*. Ed. by David W. Pearce. Cheltenham: Edward Elgar, pp. 268–285.

Posner, Eric A. and David Weisbach (2010). *Climate Change Justice*. Princeton University Press. ISBN: 9780691137759. URL: `http://www.jstor.org/stable/j.ctt7rvrw`.

Ramsey, F. P. (Dec. 1928). "A Mathematical Theory of Saving". In: *The Economic Journal* 38.152, pp. 543–559. DOI: `10.2307/2224098`. URL: `https://doi.org/10.2307/2224098`.

Richardson, Henry S. (1990). "Specifying Norms as a Way to Resolve Concrete Ethical Problems". In: *Philosophy and Public Affairs* 19.4, pp. 279–310.

Ryan, Alan (1970). *The Philosophy of the Social Sciences*. London: Macmillan.

Sachs, Jeffrey D., Andrew D. Mellinger, and John L. Gallup (2000). "The Geography of Poverty and Wealth". In: *Scientific American* 284.3, 70 to 75.

Sen, Amartya (1999). *Development as Freedom*. London: Anchor Books.

Stern, Nicholas (2007). *The Economics of Climate Change: The Stern Review*. Cambridge University Press.

World Bank (2007). *World Development Report 2008: Agriculture for Development*. Washington DC: The World Bank.

Yamaguchi, Ryo (2019). "Intergenerational Discounting with Intragenerational Inequality in Consumption and the Environment". In: *Environmental and Resource Economics* 73.4, pp. 957–972. DOI: `10.1007/s10640-018-0282-4`. URL: `https://doi.org/10.1007/s10640-018-0282-4`.